

Groups and Lie Algebras: Exercise Sheet 11

1 Highest weight modules are unique Generalizing the proof for $\mathfrak{sl}(3)$ show that for every simple Lie algebra two highest weight modules with the same highest weight are isomorphic.

2. Representations and tensor decompositions for the algebra of type $B_2 \cong C_2$ ($\mathfrak{so}(5) \cong \mathfrak{sp}(4)$)

Throughout, $V = \mathbb{C}^4$ is the defining representation of $\mathfrak{sp}(4)$, equipped with the invariant symplectic form $\Omega \in \Lambda^2 V^*$ and the invariant volume form $\text{vol} = \Omega \wedge \Omega \in \Lambda^4 V^*$. We write **4**, **5**, **10**, **1** for the irreducibles of dimension 4, 5, 10, 1. In weight coordinates we use the orthonormal L_1, L_2 with $\omega_2 = \frac{1}{2}(L_1 + L_2)$ the highest weight of V .

4 $\otimes$ 4 from the symplectic and volume forms

- (a) The form Ω defines an invariant contraction $c: \Lambda^2 V \rightarrow \mathbb{C}$. Let $\omega \in \Lambda^2 V$ be a dual bivector, $c(\omega) \neq 0$. Show that $\Lambda^2 V = \mathbb{C}\omega \oplus W$ with $W := \ker c$ an invariant subspace, $\dim W = 5$.
- (b) The wedge product $\Lambda^2 V \times \Lambda^2 V \rightarrow \Lambda^4 V \cong \mathbb{C}$ (identified via vol) is an invariant symmetric form q . Show that $q|_W$ is nondegenerate, so that the action gives a homomorphism $\rho: \mathfrak{sp}(4) \rightarrow \mathfrak{so}(W) \cong \mathfrak{so}(5)$.
- (c) Show that ρ is an isomorphism. (*Hint: $\mathfrak{sp}(4)$ is simple and $\rho \neq 0$; then compare dimensions.*)
- (d) Show that $\text{Sym}^2 V$ is irreducible with highest weight $L_1 + L_2$, the highest root. Conclude $\text{Sym}^2 V \cong \mathfrak{sp}(4)$ is the adjoint **10**.
- (e) Assemble the above into

$$\mathbf{4} \otimes \mathbf{4} = \mathbf{10} \oplus \mathbf{5} \oplus \mathbf{1}, \quad \text{Sym}^2 V = \mathbf{10}, \quad \Lambda^2 V = \mathbf{5} \oplus \mathbf{1}.$$

Identify the singlet explicitly.

Weight diagrams

- (a) Draw the root system of B_2 in the (L_1, L_2) plane. Mark the simple roots $\alpha_1 = L_1 - L_2$ and $\alpha_2 = L_2$, the fundamental weights $\omega_1 = L_1$ and $\omega_2 = \frac{1}{2}(L_1 + L_2)$.
- (b) Draw the weight diagrams of **4** and **5**.
- (c) Draw the weight diagram of the adjoint **10**.
- (d) Overlay the sixteen weights of $\mathbf{4} \otimes \mathbf{4}$ and indicate graphically how to rediscovering the decomposition above.

- e) Decompose $\mathbf{4} \otimes \mathbf{5}$ into irreducible representations.

3. Adding matter to YM

This exercise is a follow up to Exercise 3 on the previous exercise sheet. Let (ρ, V) be a finite-dimensional representation of G , with induced Lie-algebra representation $\rho_*: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. A *matter field* is a smooth map $\phi: \mathbb{R}^4 \rightarrow V$, transforming under a gauge transformation $g: \mathbb{R}^4 \rightarrow G$ by $\phi \mapsto \phi^g = \rho(g) \phi$. Its gauge-covariant derivative is

$$D_i \phi = \partial_i \phi + \rho_*(A_i) \phi.$$

- a) Show that $D_i \phi$ transforms like ϕ itself, i.e. $(D_i \phi)^g = \rho(g) D_i \phi$. (You will need $\rho_*(\text{Ad}_g X) = \rho(g) \rho_*(X) \rho(g)^{-1}$ and $\rho_*((\partial_i g)g^{-1}) = (\partial_i \rho(g)) \rho(g)^{-1}$.)
- b) Suppose V carries a G -invariant Hermitian inner product $\langle \cdot, \cdot \rangle$ (such a form exists whenever G is compact). Show that

$$S_{\text{mat}}[A, \phi] = \int_{\mathbb{R}^4} \left(\sum_i \langle D_i \phi, D_i \phi \rangle + m^2 \langle \phi, \phi \rangle \right) d^4 x$$

is gauge invariant, and hence so is the total action $S_{\text{YM}} + S_{\text{mat}}$.

- c) Specialize to $G = SU(2)$ with $V = \mathbb{C}^2$ the fundamental representation (a gauged scalar doublet): write $D_i \phi$ and $\langle D_i \phi, D_i \phi \rangle$ in components. What does the construction reduce to when instead $V = \mathfrak{g}$ and $\rho = \text{Ad}$?